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Certifications:

We Can Even CREATE OUR OWN MEDICAL DEVICE And TEST IT ON OUR NEIGHBOUR

With connected devices being all the rage, it is more important than ever to ensure that the devices follow regulations and are certified. This ensures that these devices, as TUV Rheinland puts it, do not disturb and do not get disturbed. Kalyan Varma, vice president - Business Stream Products at TUV Rheinland - TUV Rheinland (India) Pvt Ltd, speaks with Dilin Anand from EFY

Q. Which are the common protocols that IoT developers seem to prefer?

A. Wi-Fi, ZigBee and Bluetooth Low Energy (BLE) are some known protocols. Apart from these, there are new protocols like IP500 by IP500 Alliance, Thread by Thread Group and others.

IP500 is becoming a big force in Europe. It is related to monitoring and sensing devices as well as building automation. It allows you to plug-and-play the sensor anywhere without any wiring and is manufacturer-neutral, too.

Q. What are the new technologies coming into this space in India?

A. New technologies like Wi-sun, OpenArea and KNX are betting on the IoT market. KNX has done a lot of work in India. Wi-sun has also had a lot of interest in setting up workshops as well as interest groups. KNX predominantly looks at lighting and lighting automation, while Wi-sun is an advanced Wi-Fi communications channel.

Q. What are the international regulations pertaining to IoT devices?

A. Most countries have local regulations but, for more uniform adoption, some also adopt regulations based on European norms or USA based FCC standards. From an IoT perspective, regulations can be split into four major aspects: wireless,

electrical safety, electromagnetic compatibility and material compliance.

European directives mandate compliance to all the above for IoT, while for North American market, electrical safety is voluntary and market-driven.

Q. Are these regulations related to the protocols used?

A. No! These regulations are based on the frequency of transmission. Local developers and manufacturers working in the field usually do not have any idea about local or international regulations in wireless.

Q. Do regulations also apply to whitespace band?

A. Different countries have different regulations. In some cases, it is licence-free but still regulated for products operating on these frequencies and in others, it is a licenced band.

Q. What is the main challenge for IoT products?

A. If you look at protocols, the big thing to note is the wireless communications channels or frequency these prefer. Most choose to function in 2.4GHz or 5GHz. This band is also known as the industrial, scientific and medical (ISM) radio band. The moment you touch a frequency for communications, every country has a regulatory barrier in place. The challenge is to build your product such that it follows local and inter-

national regulations both.

Q. What are the challenges faced by engineering teams in India?

A. If you take a Bluetooth module, you can only make a few changes keeping in with regulations. Otherwise you will have to rework the entire design and redo the certification. Major intellectual property suppliers who do this are not available in India. Getting compliance in India and finding people who have the expertise to understand and implement the solutions are very rare here. They need to be first looked at in the form of regulatory qualifications and then protocol qualifications.

Q. How do things work for IoT health devices where there could be a mix of medical and wireless regulations?

A. We do not have a regulation set up yet in India for medical devices. As per official data, we only have a total of 38 medical devices that are regulated, and all of these are non-electrical.

Equipment from incubators to computed tomography scan machines, everything is non-regulated. But, use of radioactive materials in medical devices is regulated by AERB.

In India, we can even create our own medical device and test it on our neighbour. A bill proposed to solve this is due in the Parliament for approval. Central Drugs Standard Control Organisation is also responsible for some level of device regulation in India. **EFY**